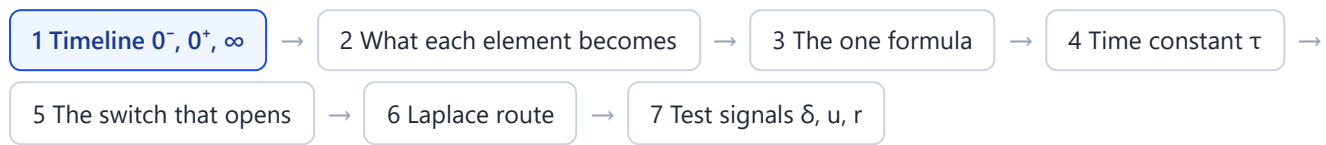


08 — First-order transients

Class notes p.19–23, set 2 p.15–22 · MTE syllabus item 6 (step-wise, Laplace, test signals) · 4 board questions + 2 textbook · needs 01 (what can't jump), 06 (R_{Th} for the time constant)

Map



The questions this file answers

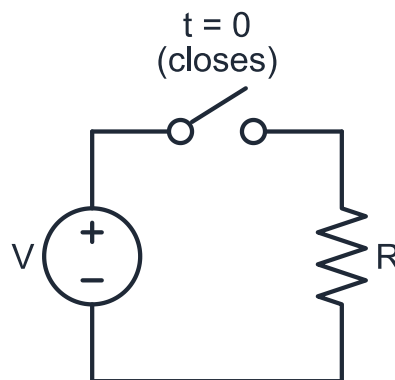
- **Class p.19–20** V , a switch that closes at $t = 0$, R : find I at 0^- , 0^+ , ∞ .
- **Class p.21** V , switch, R , L in series; the switch closes at $t = 0$: find I_R and I_L at 0^- , 0^+ , ∞ .
- **Set 2 p.20–21** V , switch (closes at $t = 0$), R , C in series: find v_C , v_R and I at 0^- , 0^+ , ∞ .
- **Class p.22–23 · set 2 p.18–19** 10 V, switch (opens at $t = 0$), 1 k Ω , 1 mH: find I_R and I_L at 0^- and 0^+ .
- **textbook** The class RL circuit solved by Laplace; RC responses to $\delta(t)$, $u(t)$, $r(t)$.

Build

1 · The timeline

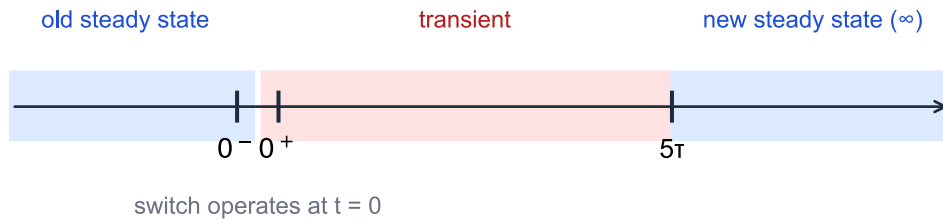
CLASS P.19–20 · SET 2 P.15–16

V , a switch that closes at $t = 0$, and R . Find $I(0^-)$, $I(0^+)$ and $I(\infty)$.



GUESS FIRST

Is there any "transient" at all in this circuit, i.e. any gradual change?



Before the switch: the old steady state. From 0^+ to about 5τ : the transient. After that: the new steady state.

- $t = 0^-$: the instant **just before** the switch operates. Old steady state.
- $t = 0^+$: the instant **just after**.
- $t = \infty$: long after (in practice about 5τ): new steady state.

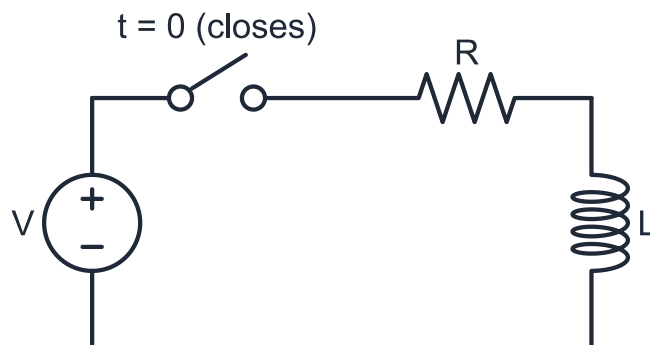
For the resistor-only circuit: $I(0^-) = 0$ (switch open), $I(0^+) = V/R$, $I(\infty) = V/R$. The current jumps straight to its final value, because a resistor has no memory (class p.20: "a resistor allows sudden change").

So the Guess answer: **no transient**. A transient needs an element that stores energy, an L or a C.

2 · What each element becomes at 0^+ and at ∞

CLASS P.21 · SET 2 P.17

V , switch, R and L in series; the switch closes at $t = 0$. Find I_R and I_L at 0^- , 0^+ and ∞ .

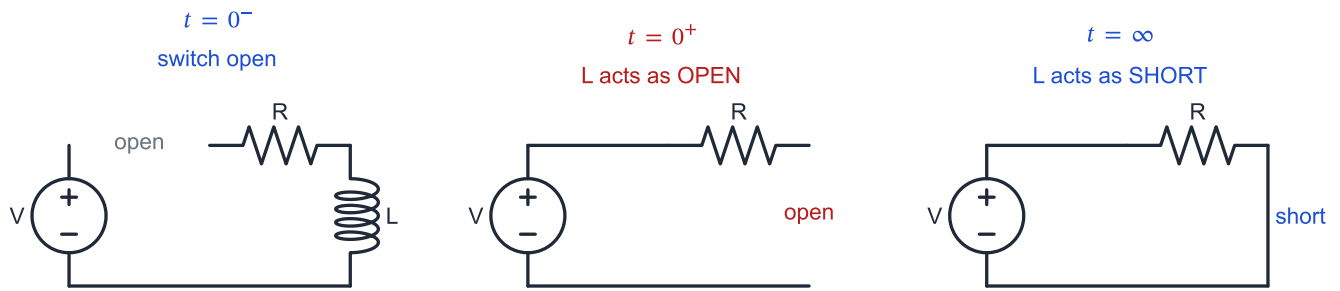


GUESS FIRST

The switch has just closed. The source is pushing. Why can't the current be V/R at $t = 0^+$?

Two rules do all the work (from 01, step 5):

Element	At 0^+ : it keeps its 0^- value	At ∞ (DC steady state)
Inductor	a current source of value $i_L(0^-)$; an open circuit if that value is 0	short circuit
Capacitor	a voltage source of value $v_C(0^-)$; a short circuit if that value is 0	open circuit
Resistor	no memory: whatever the rest of the circuit forces	same

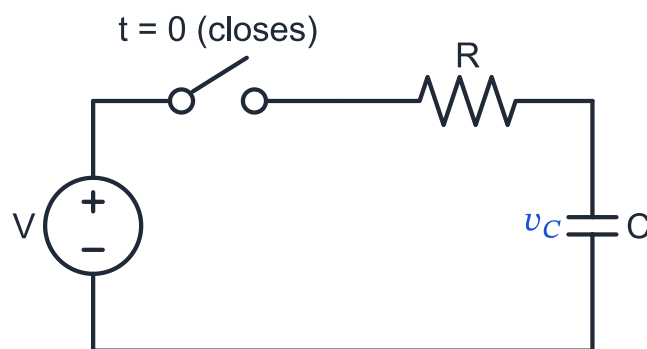


- 0^- : switch open, so $I_R(0^-) = I_L(0^-) = 0$.
- 0^+ : i_L can't jump, so $I_L(0^+) = 0$ and the inductor acts as an **open**. In series, $I_R(0^+) = 0$.
- ∞ : the inductor is a **short**, so $I(\infty) = V/R$.

The current must travel from 0 to V/R . How it travels is step 3.

SET 2 P.20-21

V , a switch that closes at $t = 0$, R and C in series (C uncharged). Find v_C , v_R and I at 0^- , 0^+ and ∞ .



The mirror image of the inductor:

- 0^- : switch open, steady state: $v_C(0^-) = 0$, $v_R(0^-) = 0$, $I = 0$.
- 0^+ : $v_C(0^+) = v_C(0^-) = 0$. The class writes it as the rule to remember: " $v_C(0^-) = v_C(0^+)$, always". So the capacitor acts as a **short**, the whole V is across R : $v_R(0^+) = V$, $I(0^+) = V/R$. The current jumps; the capacitor voltage does not.
- ∞ : $I = C \cdot \frac{dv_C}{dt} = 0$ because v_C is constant, so the capacitor is an **open** (set 2 p.21): $I(\infty) = 0$, $v_C(\infty) = V$, $v_R(\infty) = 0$.

CHECK 1

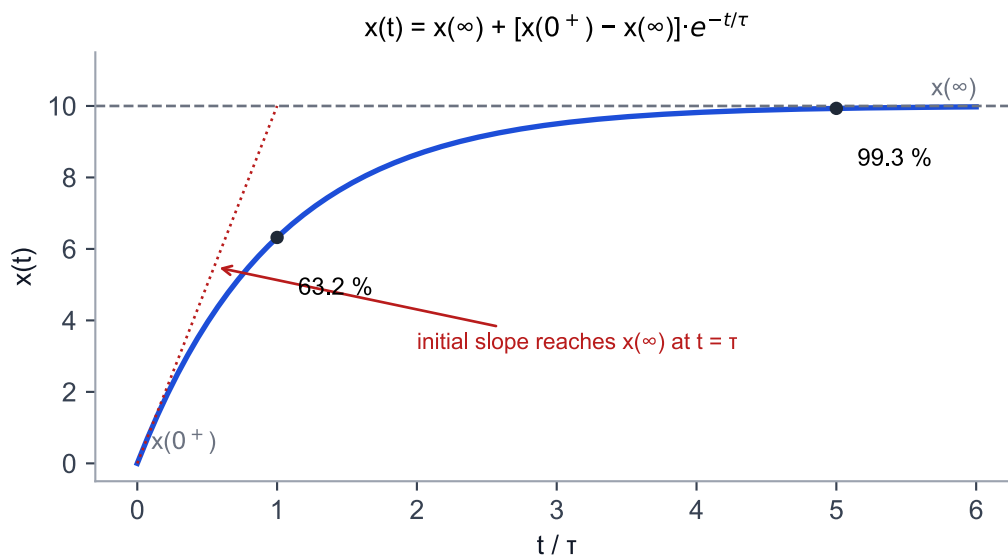
Same RC circuit, but C was charged to 4 V (+ at the top) before $t = 0$, with $V = 10$ V and $R = 1$ k Ω . Find $I(0^+)$ and $I(\infty)$.

3 · The one formula for every first-order circuit

Every circuit with **one** L or **one** C (after reduction) moves from its 0^+ value to its ∞ value **exponentially**:

$$x(t) = x(\infty) + [x(0^+) - x(\infty)] \cdot e^{-t/\tau}$$

The class p.22 box: valid for **RL ✓ and RC ✓**, not for **RLC X or LC X** (those are second order: they can oscillate).



After one τ the change is 63.2 % complete; after 5τ , 99.3 %. The initial slope would reach $x(\infty)$ in exactly one τ .

So a first-order question is always **three numbers**: $x(0^+)$, $x(\infty)$, τ . Then write the formula.

4 • The time constant τ

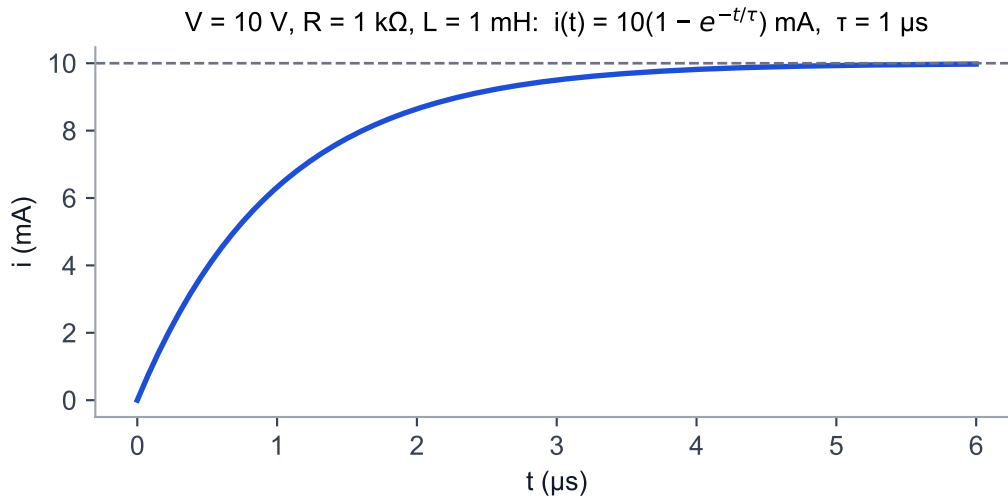
$$\tau = \frac{L}{R} \quad (\text{RL}) \quad \tau = RC \quad (\text{RC})$$

R is the Thévenin resistance seen by the L or C (06), with independent sources switched off. It is the plain R only when the L or C sees a single resistor.

Class p.21 circuit with the p.22 values ($V = 10 \text{ V}$, $R = 1 \text{ k}\Omega$, $L = 1 \text{ mH}$): $x(0^+) = 0$, $x(\infty) = 10 \text{ mA}$, $\tau = \frac{1 \text{ mH}}{1 \text{ k}\Omega} = 1$

μs :

$$i(t) = 10(1 - e^{-t/1 \mu\text{s}}) \text{ mA}$$



Confirmed by simulating the circuit step by step: at $t = 3 \mu\text{s}$ both give 9.50 mA.

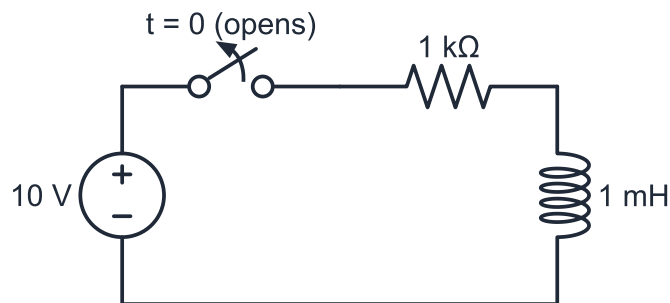
CHECK 2

An RC circuit: $R = 2 \text{ k}\Omega, C = 5 \mu\text{F}$. Find τ , and the time after which it is "settled" (99 %).

5 • The switch that opens

CLASS P.22–23 • SET 2 P.18–19

10 V, a switch that opens at $t = 0$, $1 \text{ k}\Omega$ and 1 mH in series. Find I_R and I_L at 0^- and 0^+ .



GUESS FIRST

At 0^- the loop is carrying current. At 0^+ the switch is open. What must the inductor do, and where can its current go?

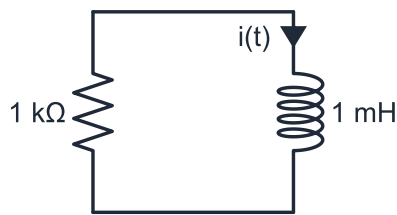
- 0^- : switch closed, steady state, $L = \text{short}$: $I_L(0^-) = 10 \text{ V} / 1 \text{ k}\Omega = 10 \text{ mA} = I_R(0^-)$.
- 0^+ : the inductor current can't jump, so $I_L(0^+) = 10 \text{ mA} \checkmark$ (the professor's point).

READ THIS BEFORE WRITING $I_R(0^+) = 0$

Both sets of notes give $I_R(0^+) = 0$ alongside $I_L(0^+) = 10 \text{ mA}$ (set 2 p.19 draws the inductor as a 10 mA source). R and L are **in the same series loop**, so they must carry the **same** current. Both statements can't hold. With the switch open there is no path at all for the inductor's 10 mA. In an ideal circuit, $di/dt \rightarrow \infty$ and the inductor voltage spikes. In a real one, the stored energy $\frac{1}{2}Li^2 = \frac{1}{2} \times 1 \text{ mH} \times (10 \text{ mA})^2 = 50 \text{ nJ}$ arcs across the opening contacts.

Ask the professor which path the current takes after $t = 0$. Usually the full question has the switch move the R–L onto its own loop, or has a resistor across the switch.

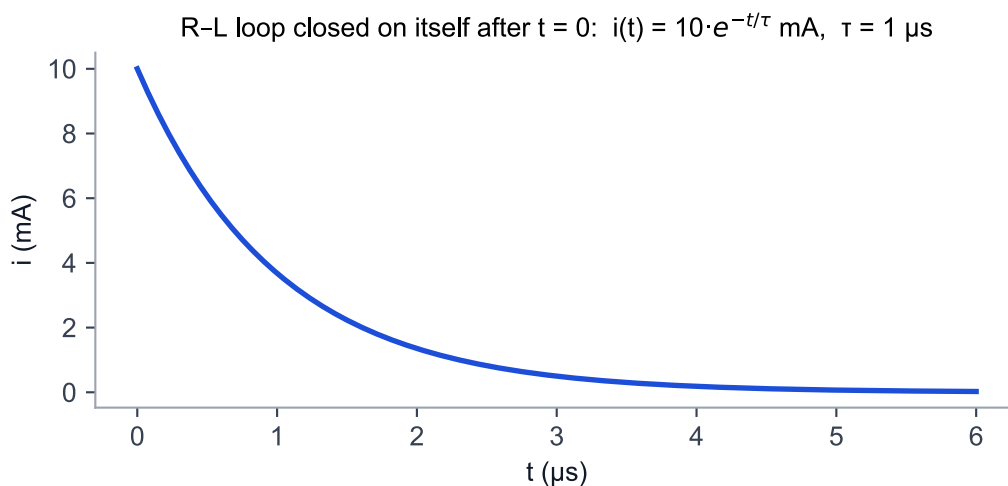
If the R–L loop is closed on itself after $t = 0$ (the usual version of this question):



Illustration, not the board circuit: the R–L loop with the source gone.

$x(0^+) = 10 \text{ mA}$, $x(\infty) = 0$, $\tau = L/R = 1 \mu\text{s}$:

$$i(t) = 10 \cdot e^{-t/1 \mu\text{s}} \text{ mA}$$



6 · The Laplace route

TEXTBOOK

Solve the class p.21 circuit (10 V switched onto 1 kΩ and 1 mH at $t = 0$) by Laplace transform.

GUESS FIRST

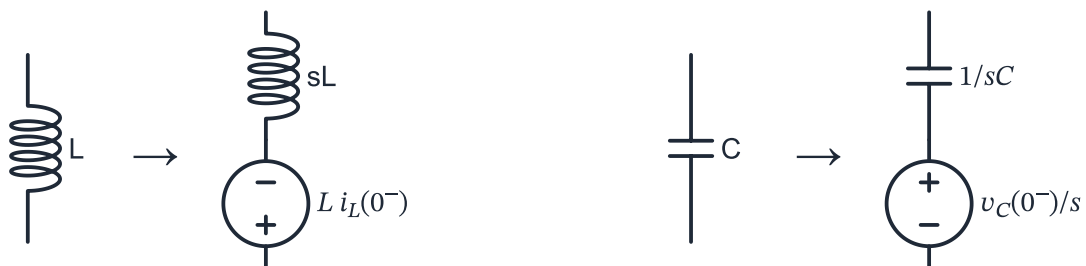
The answer is already known from step 4. What should the Laplace working give back, and at which step does the "1 - e" shape appear?

The step-wise method needs the exponential shape to be known in advance. Laplace **derives** it: it turns d/dt into multiplication by s, so the circuit's differential equation becomes algebra.

Pairs you need (the 06-KB table; the full list is in step 7):

f(t)	F(s)
$\delta(t)$	1
u(t) (a DC source switched on at 0)	$\frac{1}{s}$
t·u(t)	$\frac{1}{s^2}$
$e^{-at}u(t)$	$\frac{1}{s+a}$
$\frac{df}{dt}$	$sF(s) - f(0^-)$

Elements in the s-domain. $R \rightarrow R$. $L \rightarrow sL$. $C \rightarrow \frac{1}{sC}$. Energy stored at $t = 0^-$ becomes a source in series:



Current enters the top terminal. The inductor's initial current appears as a voltage source $L \cdot i_L(0^-)$; the capacitor's initial voltage as $v_C(0^-)/s$.

The class circuit. $i_L(0^-) = 0$, so no initial source. The 10 V switched on at $t = 0$ is $10 \cdot u(t) \rightarrow \frac{10}{s}$. KVL in s:

$$\frac{10}{s} = (R + sL) \cdot I(s) \quad \Rightarrow \quad I(s) = \frac{10}{s(1000 + 0.001s)} = \frac{10^4}{s(s + 10^6)}$$

Partial fractions: $\frac{10^4}{s(s + 10^6)} = \frac{A}{s} + \frac{B}{s + 10^6}$, with $A = \frac{10^4}{10^6} = 0.01$ and $B = -0.01$:

$$I(s) = \frac{0.01}{s} - \frac{0.01}{s + 10^6} \quad \Rightarrow \quad i(t) = 0.01(1 - e^{-10^6 t}) \text{ A} = 10(1 - e^{-t/1 \mu\text{s}}) \text{ mA}$$

The same answer as step 4. The "1 - e" comes from the two partial fractions: $1/s$ gives the final value, $1/(s + 1/\tau)$ gives the decaying part. The pole at $s = -10^6 = -\frac{1}{\tau}$ is the time constant.

With stored energy. The step 5 decay (R-L loop, $i_L(0^-) = 10$ mA, no source): KVL with the initial-current source, $0 = (R + sL)I(s) - L \cdot i_L(0^-)$, so

$$I(s) = \frac{i_L(0^-)}{s + R/L} \Rightarrow i(t) = 10e^{-t/1 \mu s} \text{ mA}$$

Quick checks without inverting (final- and initial-value theorems): $i(\infty) = \lim_{s \rightarrow 0} sI(s) = \frac{10^4}{10^6} = 10 \text{ mA} \checkmark$; $i(0^+) = \lim_{s \rightarrow \infty} sI(s) = 0 \checkmark$. The final-value theorem only holds if the response settles (no poles on the $j\omega$ axis or to the right).

CHECK 3

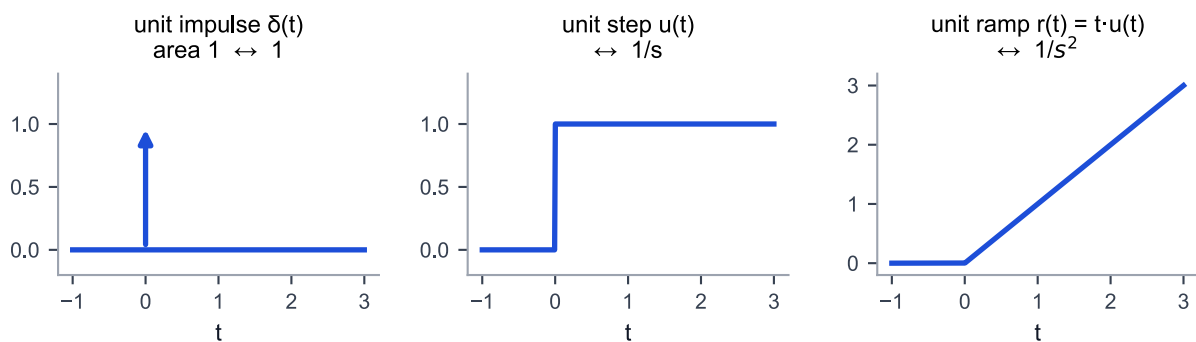
Write $I(s)$ for 10 V switched onto $R = 2 \text{ k}\Omega$ and $C = 1 \mu\text{F}$ in series (C uncharged), and invert it.

7 · Standard test signals: $\delta(t)$, $u(t)$, $r(t)$

PREDICT

A step is the integral of an impulse. If you know how a circuit responds to an impulse, how could you get its step response without solving again?

The class list (set 2 p.22), with its everyday examples:



The three test signals and their Laplace transforms.

Signal	Definition	$\mathcal{L}$	Class example	Tests...
unit impulse $\delta(t)$	zero except at $t = 0$; area $\int \delta(t) dt = 1$	1	a kitchen lighter's spark, a bike's spark plug	the response to a sudden kick
unit step $u(t)$	0 for $t < 0$, 1 for $t > 0$	$\frac{1}{s}$	a switch closing	the response to a sudden constant input
unit ramp $r(t)$	$t \cdot u(t)$	$\frac{1}{s^2}$	pressing an accelerator	tracking a steadily rising input: the "toughest" test signal (class)

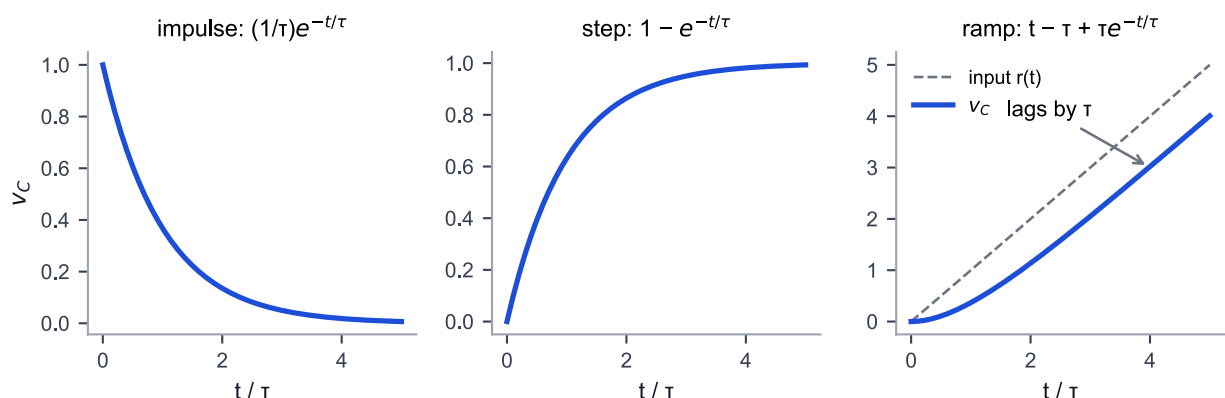
They are linked: $\delta(t) = \frac{du}{dt}$ and $u(t) = \frac{dr}{dt}$. So in a linear circuit **the step response is the integral of the impulse response**, and the ramp response is the integral of the step response. The Predict answer: integrate.

TEXTBOOK

RC circuit: $R = 1 \text{ k}\Omega$, $C = 1 \text{ }\mu\text{F}$ ($\tau = RC = 1 \text{ ms}$), initially uncharged, output v_C . Find $v_C(t)$ for a unit impulse, a unit step and a unit ramp at the input.

Voltage divider in s : $V_C(s) = V_{in}(s) \cdot \frac{1/sC}{R + 1/sC} = V_{in}(s) \cdot \frac{1}{1 + s\tau}$.

Input	$V_C(s)$	Partial fractions	$v_C(t)$
$\delta(t)$	$\frac{1}{1 + s\tau}$	$\frac{1/\tau}{s + 1/\tau}$	$\frac{1}{\tau} e^{-t/\tau}$
$u(t)$	$\frac{1}{s(1 + s\tau)}$	$\frac{1}{s} - \frac{1}{s + 1/\tau}$	$1 - e^{-t/\tau}$
$r(t)$	$\frac{1}{s^2(1 + s\tau)}$	$\frac{1}{s^2} - \frac{\tau}{s} + \frac{\tau}{s + 1/\tau}$	$t - \tau + \tau e^{-t/\tau}$



All three checked by simulating the circuit: at $t = 1 \text{ ms}$ and 3 ms the simulated and formula values agree.

Read the ramp result: after the transient dies, $v_C = t - \tau$. The output follows the input with a **constant lag of $\tau = 1 \text{ ms}$** that never goes away. That permanent error is why the ramp is called the toughest test.

CHECK 4

Differentiate the step response $1 - e^{-t/\tau}$. Which response do you get?

Exam form

Procedure for any first-order switching question. 1. $t = 0^-$: old circuit, DC steady state ($L = \text{short}$, $C = \text{open}$). Find $i_L(0^-)$ or $v_C(0^-)$. 2. $t = 0^+$: new circuit. $L \rightarrow \text{current source } i_L(0^-)$; $C \rightarrow \text{voltage source } v_C(0^-)$. Find the asked quantity. 3. $t = \infty$: new circuit, DC steady state ($L = \text{short}$, $C = \text{open}$). 4. $\tau = L/R_{Th}$ or $R_{Th}C$, with R_{Th} seen by the L or C . 5. $x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$.

Laplace route. (1) Find $i_L(0^-)$, $v_C(0^-)$. (2) Transform: sources $\rightarrow F(s)$, $R \rightarrow R$, $L \rightarrow sL$ (+ source $L i_L(0^-)$), $C \rightarrow 1/sC$ (+ source $v_C(0^-)/s$). (3) KVL/KCL in s , solve for the unknown. (4) Partial fractions, inverse transform.

Test signals. $\delta(t) \leftrightarrow 1$, $u(t) \leftrightarrow 1/s$, $r(t) = t \cdot u(t) \leftrightarrow 1/s^2$; $\delta = du/dt$, $u = dr/dt$. RC ($\tau = RC$): impulse $(1/\tau)e^{-t/\tau}$, step $1 - e^{-t/\tau}$, ramp $t - \tau + \tau e^{-t/\tau}$.

Facts to state. Inductor current and capacitor voltage cannot change instantaneously ($v_C(0^-) = v_C(0^+)$, $i_L(0^-) = i_L(0^+)$). Settling $\approx 5\tau$ (99.3 %). 63.2 % of the change happens in one τ .

Traps

- Using the ∞ circuit at 0^+ (treating L as a short immediately).
- Calling $i_L(0^+) = 0$ when the inductor was carrying current at 0^- : it keeps its 0^- value, whatever it was.
- $\tau = L/R$ with the wrong R : it must be the Thévenin resistance the L sees.
- Using the first-order formula on an RLC circuit.
- Laplace: forgetting the initial-condition source, or writing a DC source as V instead of V/s .
- Using the final-value theorem on a response that oscillates forever.

Self-test

1. What does a capacitor look like at 0^+ if it was uncharged? And at ∞ ?
2. Why is $\tau = L/R$ and not LR ? Check with units: $H/\Omega = ?$
3. RL circuit, $i(0^+) = 2 \text{ A}$, $i(\infty) = 0$, $\tau = 5 \text{ ms}$. Write $i(t)$. What is i at 10 ms ?
4. After how many time constants is a transient 99 % done?
5. Why is there no transient in a purely resistive circuit?
6. Give the Laplace transform of $\delta(t)$, $u(t)$, $r(t)$, and one everyday example of each.
7. Find $I(s)$ and $i(t)$ for an RL circuit ($R = 1 \text{ k}\Omega$, $L = 1 \text{ H}$) with $i(0^-) = 5 \text{ mA}$ and no source.

Answers

Check 1. $v_C(0^+) = 4 \text{ V}$, so the resistor sees $10 - 4 = 6 \text{ V}$: $i(0^+) = \mathbf{6 \text{ mA}}$. At ∞ the C is open: $i(\infty) = \mathbf{0}$ ($v_C(\infty) = 10 \text{ V}$).

Check 2. $\tau = 2 \text{ k}\Omega \times 5 \text{ }\mu\text{F} = \mathbf{10 \text{ ms}}$; settled after about $5\tau = \mathbf{50 \text{ ms}}$.

Check 3. $I(s) = \frac{10/s}{2000 + 1/(10^{-6}s)} = \frac{0.005}{s + 500} \Rightarrow i(t) = \mathbf{5e^{-500t} \text{ mA}}$ ($\tau = 2 \text{ ms}$; $i(0^+) = 10 \text{ V}/2 \text{ k}\Omega \checkmark$).

Check 4. $\frac{1}{\tau}e^{-t/\tau}$: the **impulse response**, as $\delta = du/dt$ predicts.

Self-test.

1. A **short** at 0^+ ; an **open** at ∞ .
2. $H/\Omega = (V \cdot s/A)/(V/A) = \mathbf{s}$: L/R is a time. (LR has units $\Omega^2 \cdot s$, not a time.)
3. $i(t) = \mathbf{2e^{-t/5 \text{ ms}} \text{ A}}$; at 10 ms , $i = 2e^{-2} = \mathbf{0.27 \text{ A}}$.
4. About $\mathbf{5\tau}$ (99.3 %); 4.6τ gives exactly 99 %.
5. A resistor stores no energy, so nothing needs time to build up or run down: every quantity jumps straight to its final value.
6. $\mathbf{1, 1/s, 1/s^2}$. Lighter spark (impulse), switch (step), accelerator (ramp).
7. $I(s) = \frac{L i(0^-)}{R + sL} = \frac{0.005}{s + 1000} \Rightarrow i(t) = \mathbf{5e^{-1000t} \text{ mA}}$ ($\tau = 1 \text{ ms}$).